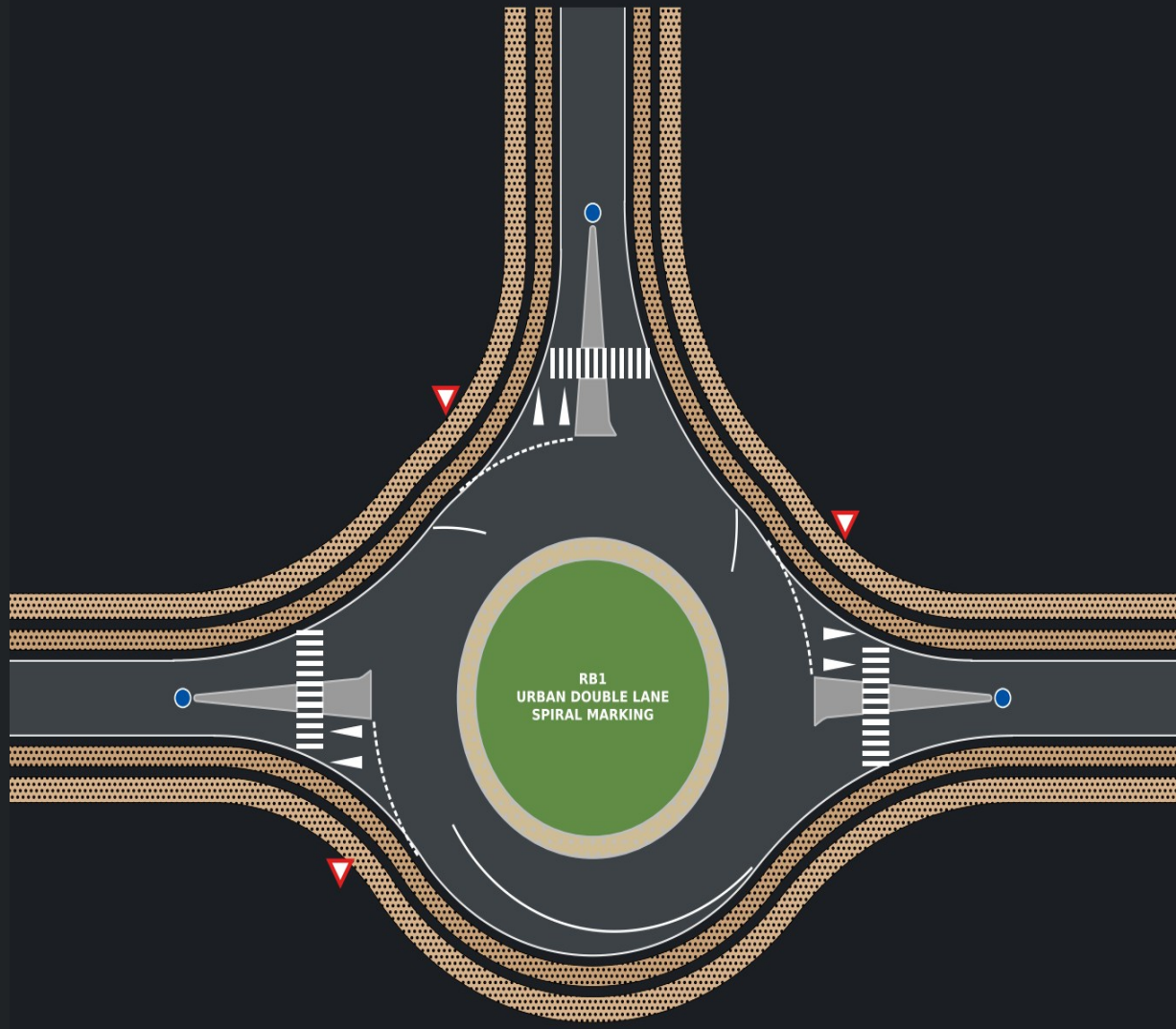
ROADPRO v3.5

Premium User Guide

Road Builder for AutoCAD and Civil 3D — every feature, step by step, with pictures



Version 3.5 · 03-Nov-2026

Developed by https://www.exceltokml.com/

What's new in v3.5

Five big additions — everything from v3.4 still works the same way.

Saudi junction standard

RDJ fills corner radius, auxiliary lane, tapers, storage and island from SHC 301 / MOMRA (AASHTO based) and draws sight triangles.

Live update

Edit a centre line with grips, MOVE, STRETCH or PEDIT — the road, corridor, widenings, junctions and roundabouts rebuild by themselves.

Corridor wraps corners

Sidewalks, walkways, drains and utility strips follow every kerb return and are hatched — no more gaps at junctions.

Roundabouts

13 design standards (FHWA 2000 and Guide 2010). Build one with RDRB, or choose “Roundabout” in RDJ and type the radius.

Concrete pavement

JPCP / JRCP with earthwork, stone base, slab, joints, dowels, tie bars and mesh — on plan, in section and in the report.

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01

Install the add-in

One double-click installs RoadPro for every AutoCAD and Civil 3D version from 2013 on. After that it loads by itself in every drawing.

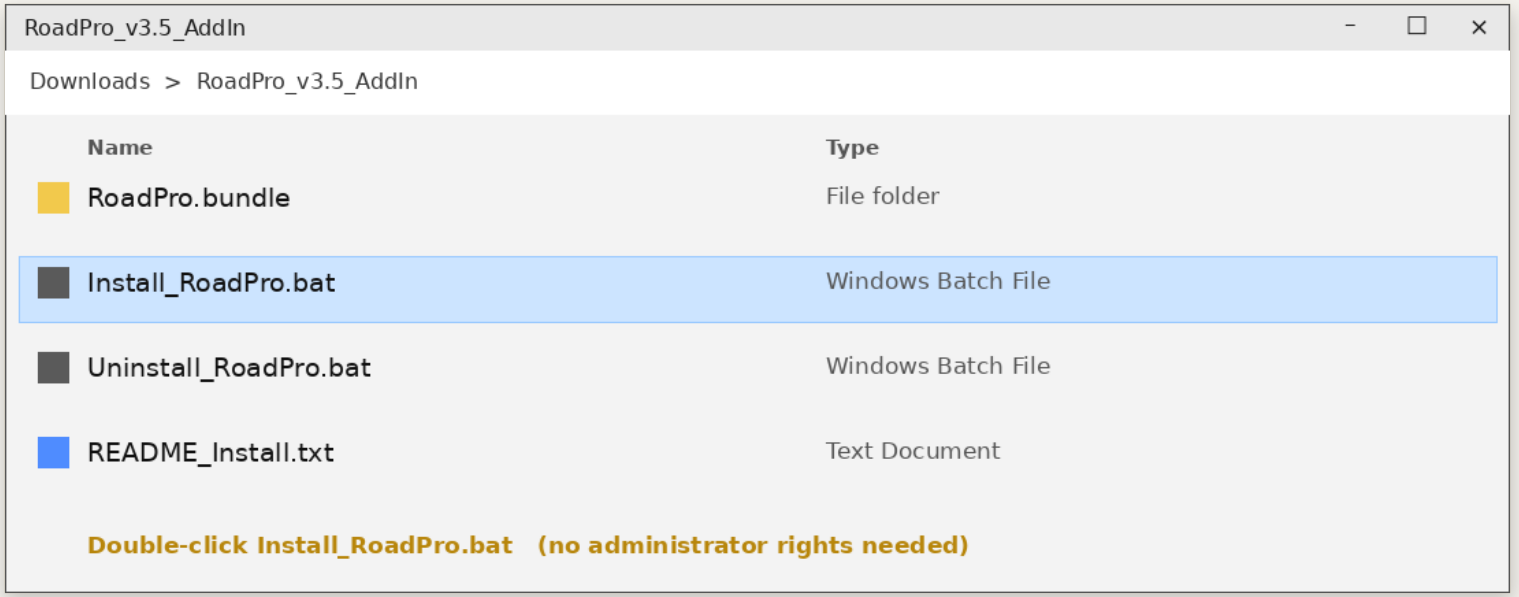
INSTALLER · RDHELP · RDINSTALL · RDUNINSTALL

1 Unzip the package and run the installer

Unzip RoadPro_v3.5_AddIn.zip to any folder. Keep RoadPro.bundle next to the .bat files.

Close AutoCAD / Civil 3D so the files are not in use.

Double-click Install_RoadPro.bat. No administrator rights are needed.



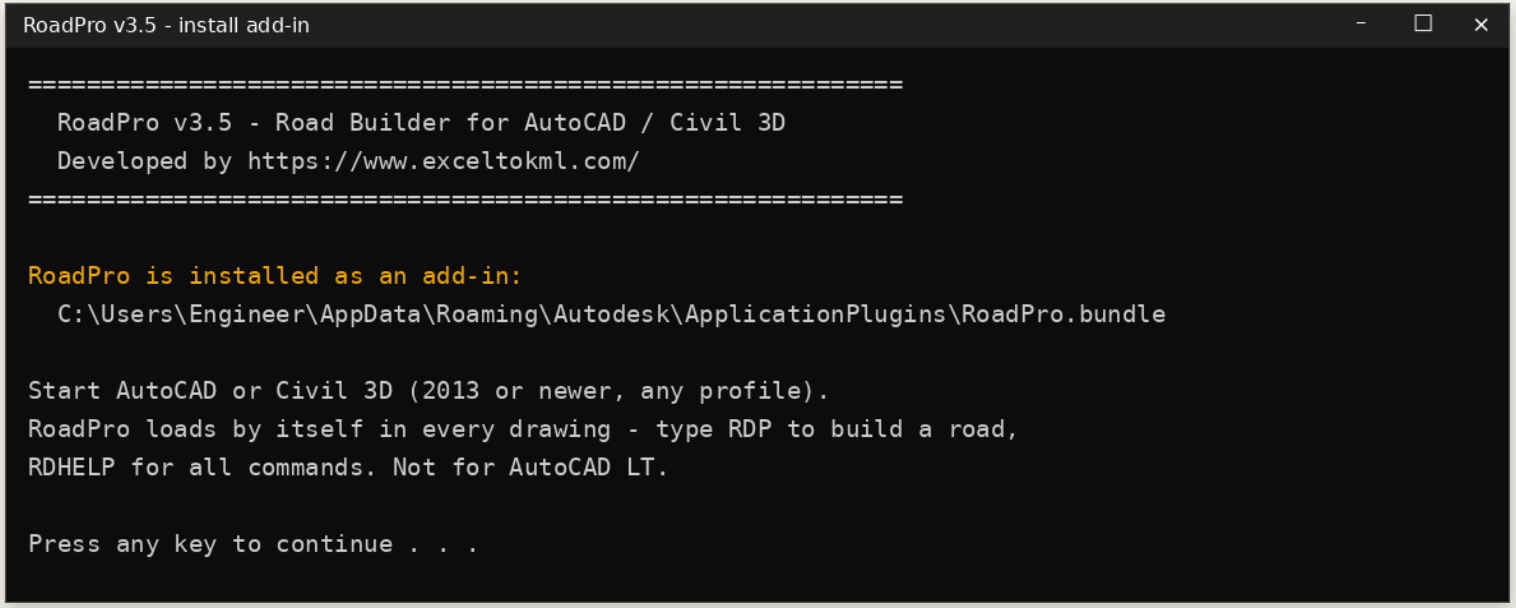
TIP The installer copies RoadPro.bundle to %APPDATA%\Autodesk\ApplicationPlugins — the official Autodesk add-in folder.

2 Read the installer message

Check the install path shown in amber. RoadPro is now an add-in.

Files are unblocked automatically, so Windows security does not stop the LISP from loading.

Press any key to close the window.



TIP Works with AutoCAD, Civil 3D, Map 3D and other AutoCAD-based products 2013 or newer. AutoCAD LT cannot run LISP add-ins.

3 Start AutoCAD — RoadPro loads by itself



1 Open any drawing

Start AutoCAD or Civil 3D as usual, in any profile (Metric or Imperial).

2 Look at the command line

“Road Pro v3.5 loaded” lists the main shortcuts and shows whether live update is ON.

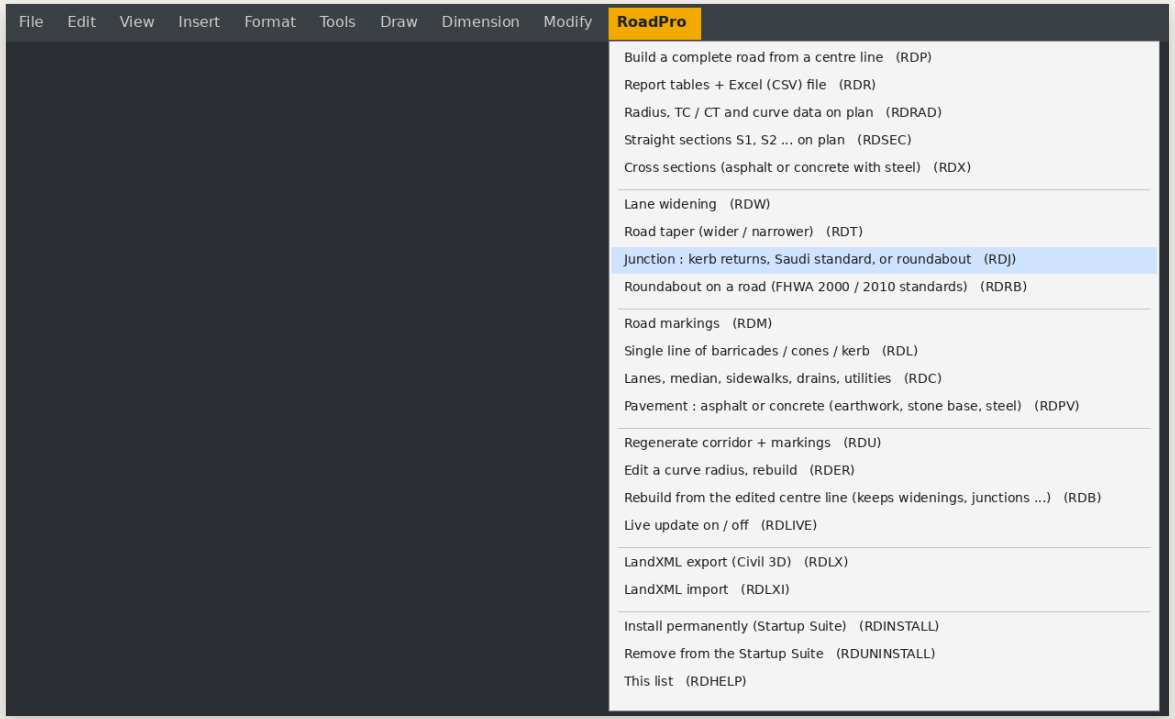
3 Start working

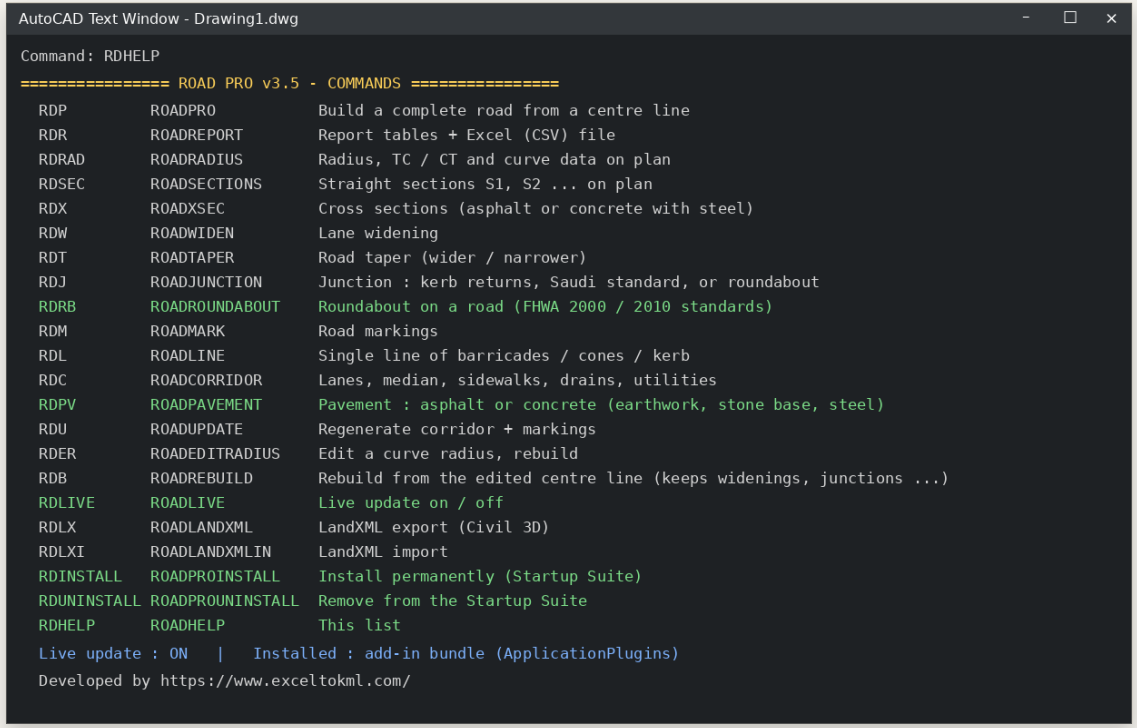
Type RDP to build your first road. RoadPro loads again in every new drawing.

! Security prompt?

If AutoCAD asks whether to load RoadPro.lsp, choose “Always load”.

4 Find every command — menu bar and RDHELP





MENUBAR = 1 shows the menu bar; the RoadPro menu is added at the end with every command and its shortcut.

Type RDHELP for the full list in the text window. New v3.5 commands are shown in green.

TIP RDHELP also tells you if live update is ON and how RoadPro is installed.

5 Other ways to install — and how to remove

A · Add-in bundle

Recommended

Install_RoadPro.bat copies RoadPro.bundle into ApplicationPlugins. Loads in every drawing, every profile, every AutoCAD version installed.

B · APPLOAD once

No installer needed

APPLOAD → RoadPro.lsp → Load. RoadPro copies itself to %APPDATA%\RoadPro and adds itself to the Startup Suite of the current profile. RDINSTALL repeats this setup.

C · Remove

Clean uninstall

Run Uninstall_RoadPro.bat for the bundle, or type RDUNINSTALL to take RoadPro out of the Startup Suite. Your drawings are not changed.

TIP Civil 3D with method B: install once in each profile you use (Metric and Imperial). The bundle (A) covers all profiles at once.

02

Build a road

One dialog sets geometry, edges, signs, markings, chainage, corridor and pavement. Then draw or pick the centre line.

RDP = ROADPRO

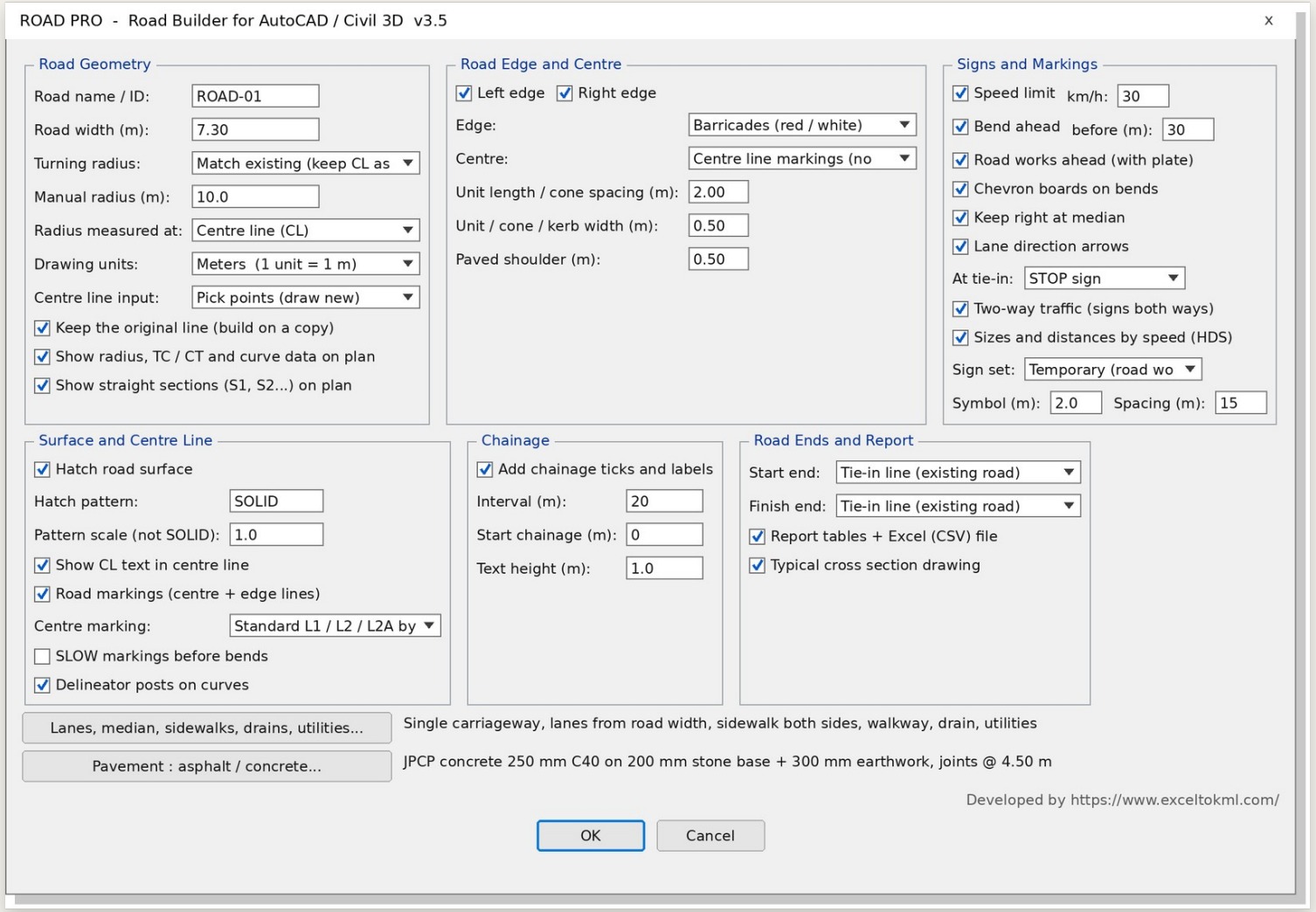
1 Type RDP to open the Road Pro dialog

Type RDP (or ROADPRO) on the command line.

Work through the six panels from top-left to bottom-right — the next steps explain each one.

Use the two buttons at the bottom for the corridor (lanes, sidewalks …) and the pavement.

Press OK, then draw or select the centre line.



TIP Your settings are remembered for the next road. ROADPRORESET restores the defaults (7.30 m wide, radius 10 m).

2 Set the road geometry

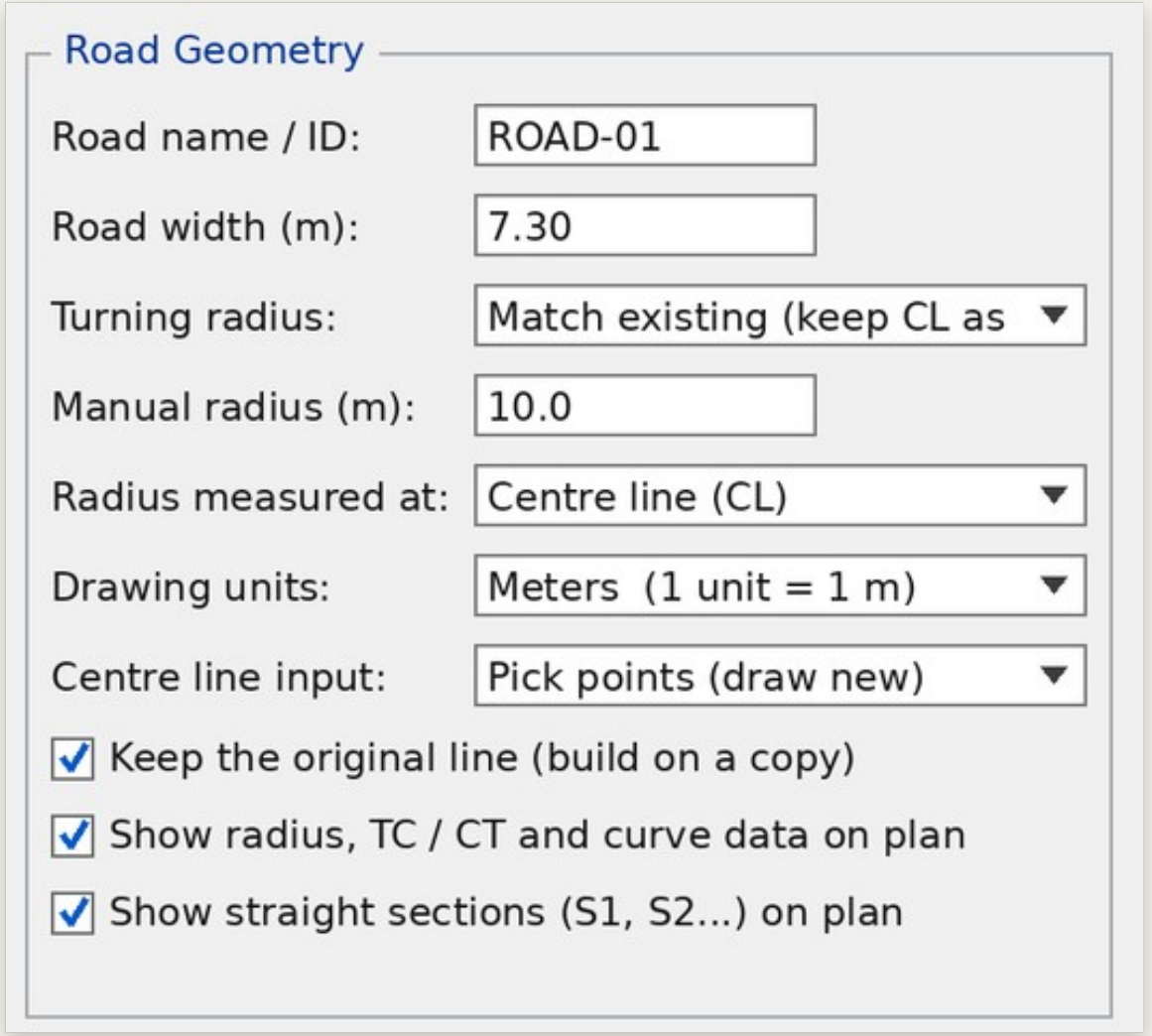
Road name and width — e.g. ROAD-01, 7.30 m.

Turning radius: match the existing bends, apply a manual radius to every bend, or none.

Radius measured at the centre line or the inner edge (kerb line).

Centre line input: pick points (draw new) or select an existing line, polyline, arc or Civil 3D alignment.

Keep the original line to build on a copy; switch on curve data and straight sections S1, S2 …



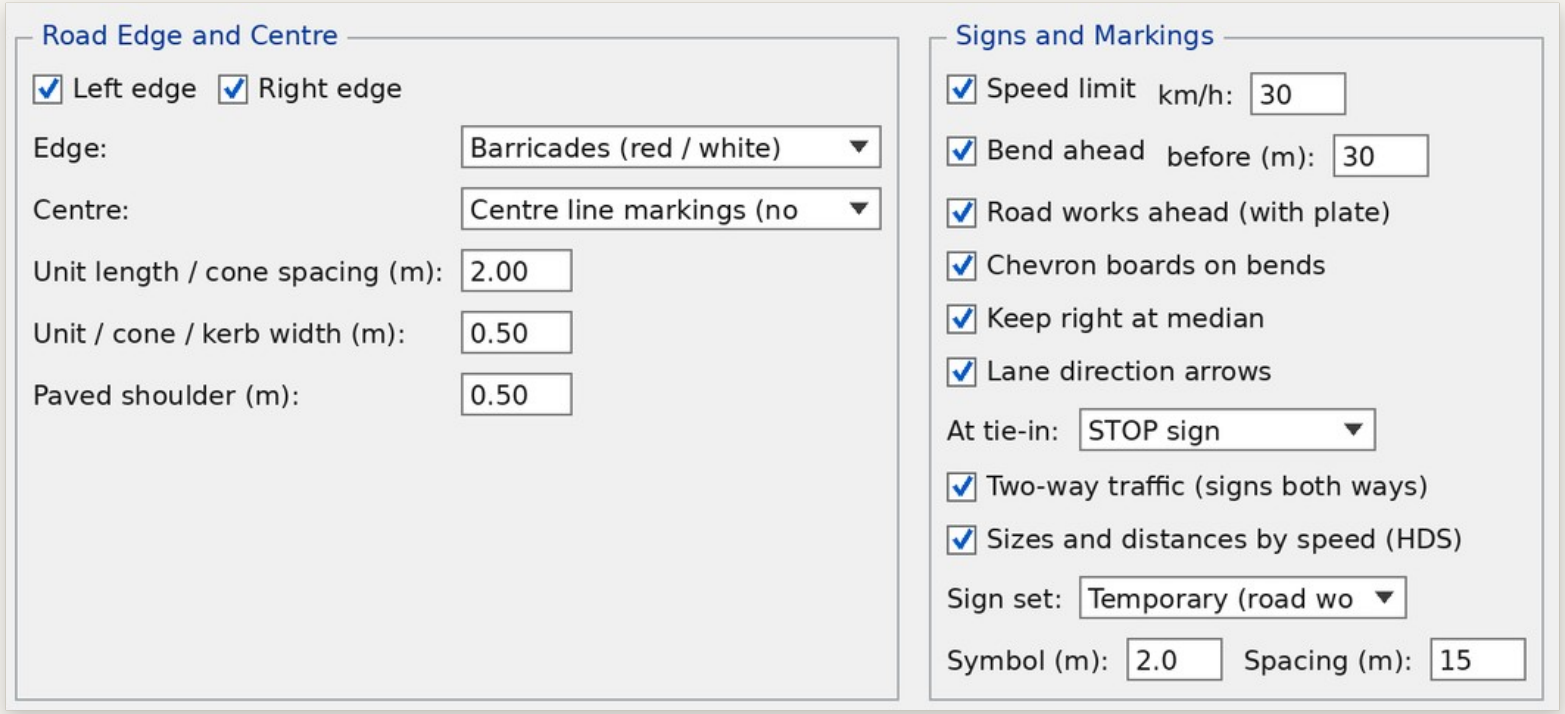
3 Choose edges, signs and markings

Edges: barricades (red / white), cones, kerb and more, left and right, with unit length and width.

Centre: centre line markings, barrier or nothing; add a paved shoulder.

Signs: speed limit, bend ahead, road works, chevrons, keep right, lane arrows, STOP / give way at the tie-in.

Sizes by speed — symbol size and spacing follow the design speed.



TIP Choose the sign set (temporary road works or permanent) before you press OK.

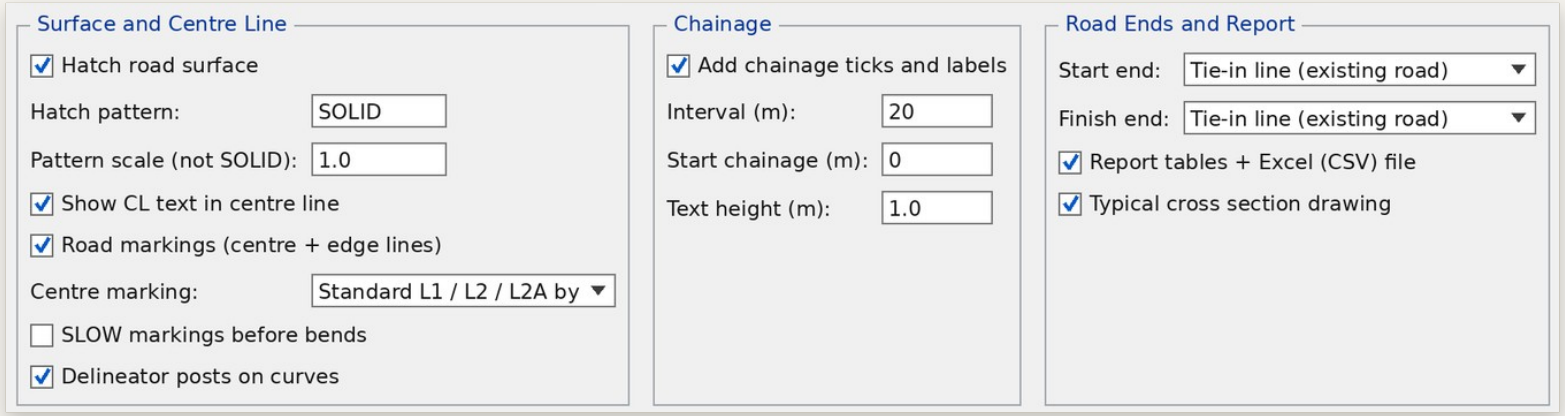
4 Surface, chainage and road ends

Hatch the road surface (SOLID or a pattern) and choose the centre marking style.

Chainage: interval, start chainage and text height.

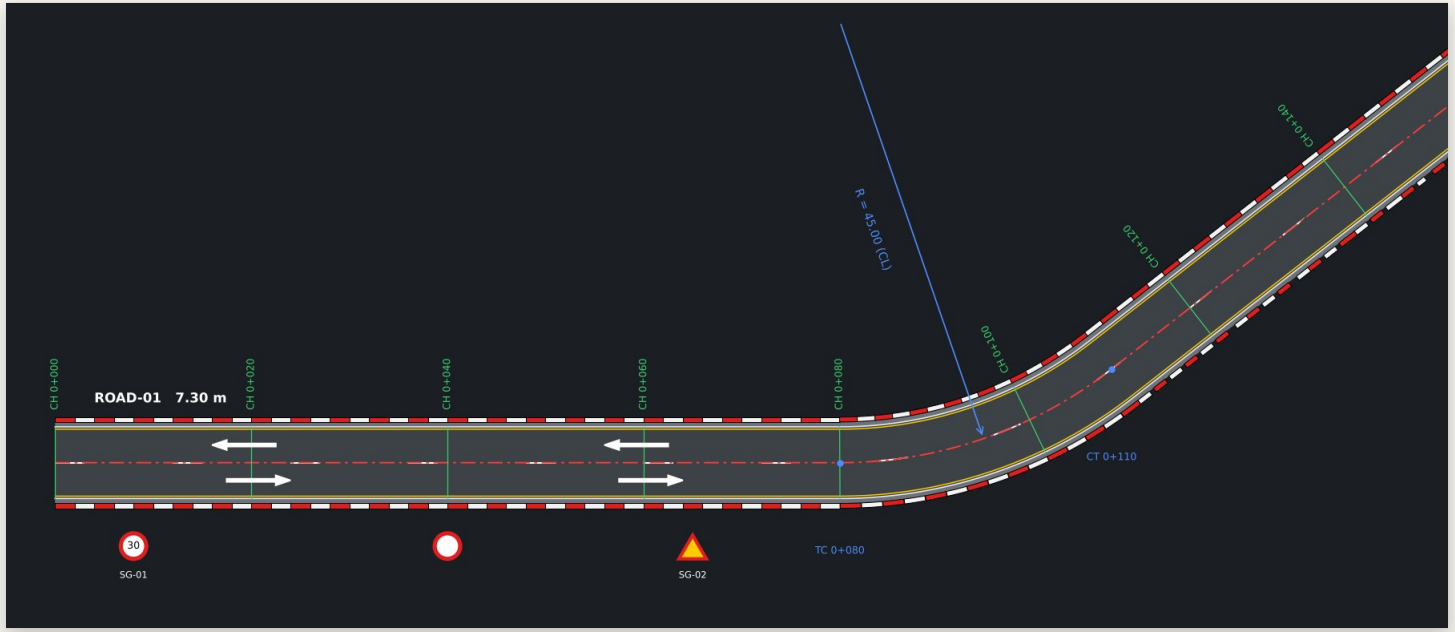
Road ends: tie-in line to an existing road, or a plain end.

Report and cross section — tick to get tables, an Excel (CSV) file and a typical section after building.



TIP Delineator posts and SLOW markings can be added on bends from this panel.

5 Draw the centre line — the road is built



1 Draw the centre line

Pick points (Arc option allowed) and press Enter — or select the existing line you chose.

2 RoadPro builds the road

Fillets every bend, draws edges, barricades, markings, chainage, signs, TC / CT and radius labels.

3 Place report and section

Pick the corner for the report tables and the point for the cross section (Enter = skip).

03

Corridor, pavement & quantities

Lanes, median, sidewalks, drains and utilities — asphalt or concrete pavement with steel — and the quantities report.

RDC · RDPV · RDX · RDR

1 Set lanes, median, sidewalks, drains and utilities

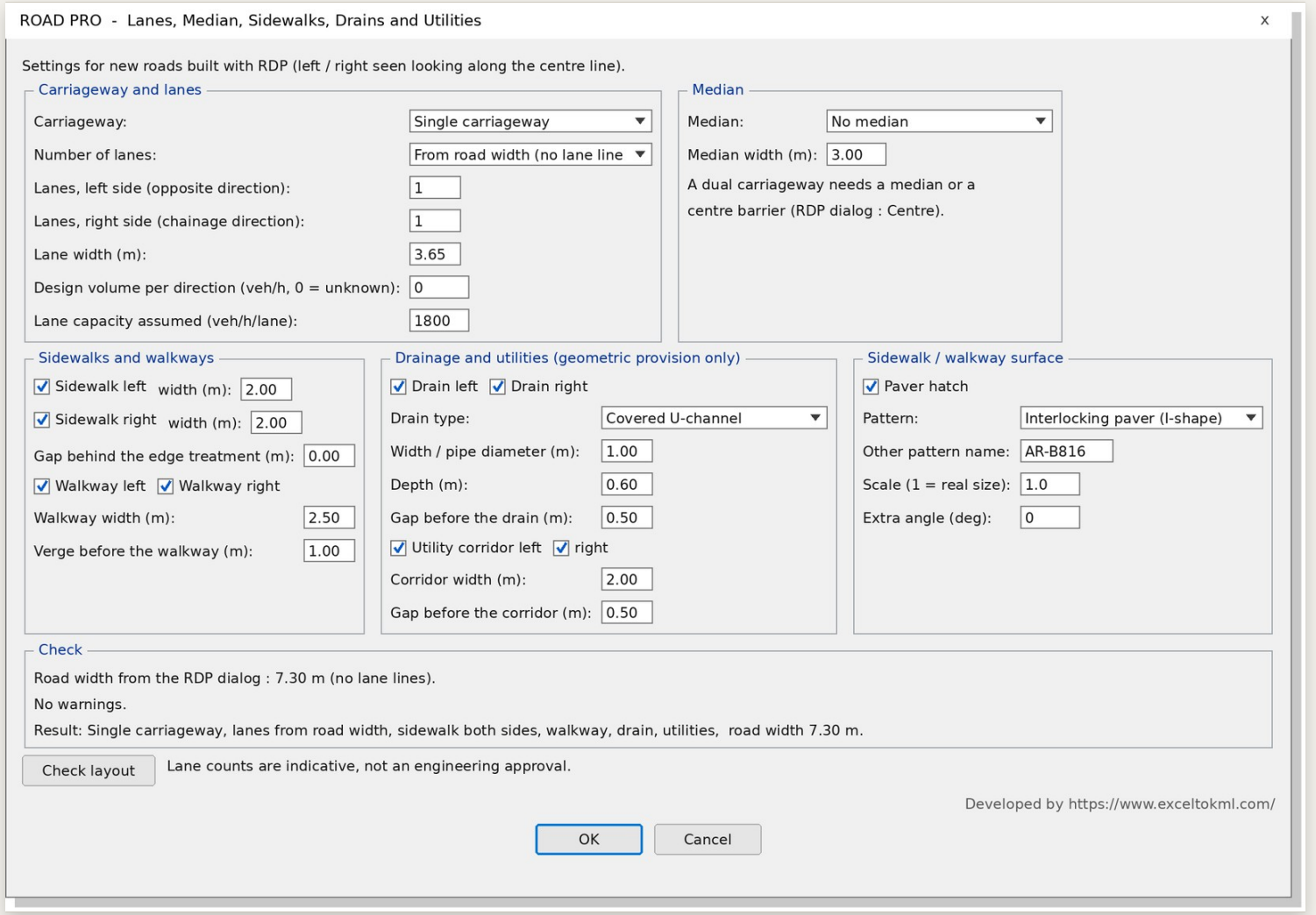
Open it from the RDP button “Lanes, median, sidewalks …” — or type RDC for a road that is already built.

Carriageway: single or dual, lanes per side and lane width.

Sidewalks and walkways left / right with widths, gap and verge.

Drains and utility corridor — covered U-channel or pipe, width, depth and gaps (geometric provision).

Paver hatch for the sidewalk surface; press Check layout.



2 Choose the pavement — asphalt or concrete

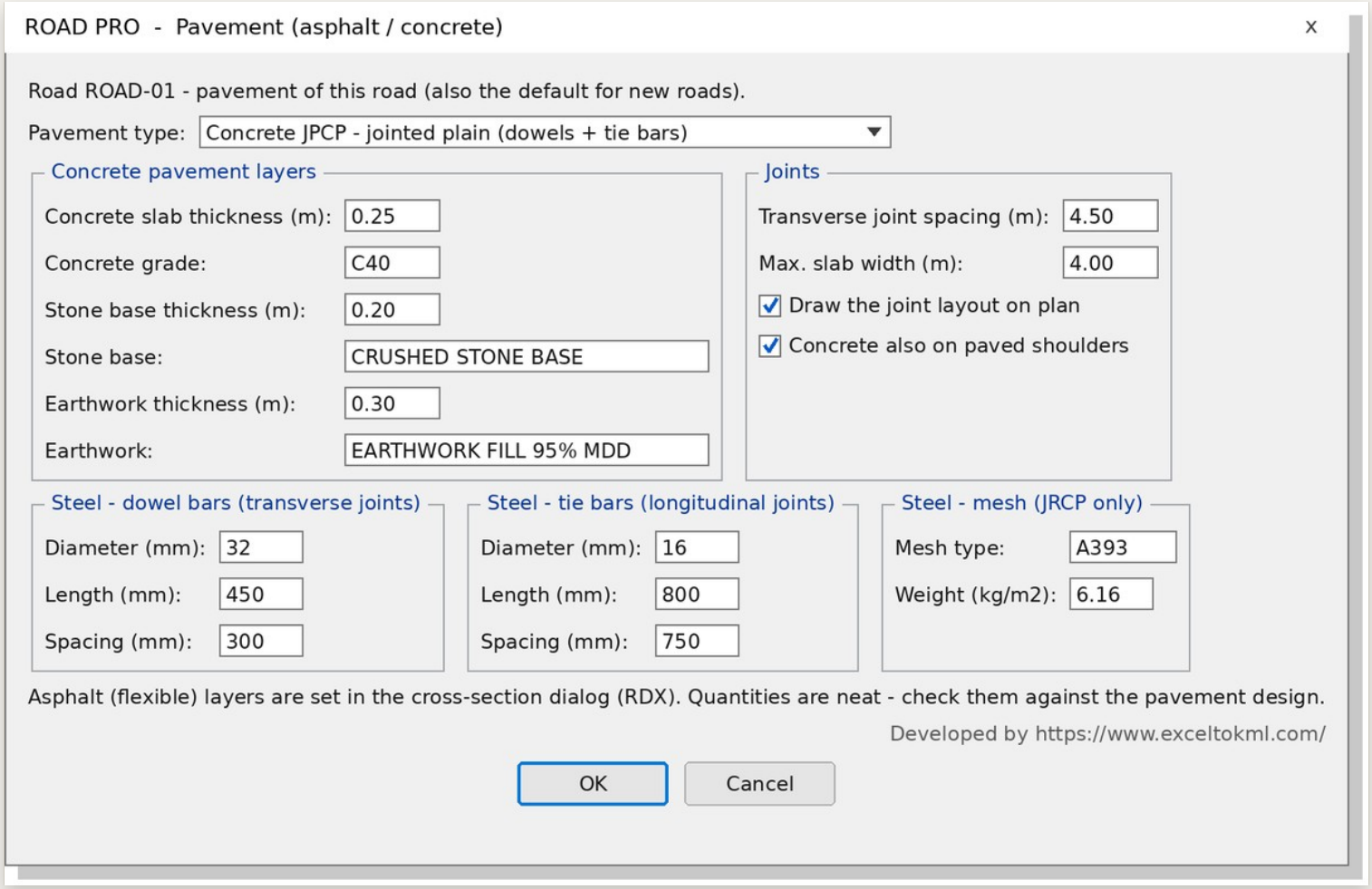
Open it from the RDP button “Pavement …” or type RDPV for a built road.

Pavement type: flexible (asphalt), concrete JPCP (dowels + tie bars) or JRCP (adds steel mesh).

Layers: concrete slab and grade, stone base and earthwork thicknesses.

Joints: transverse joint spacing and maximum slab width.

Steel: dowel bars, tie bars and mesh — diameter, length and spacing.

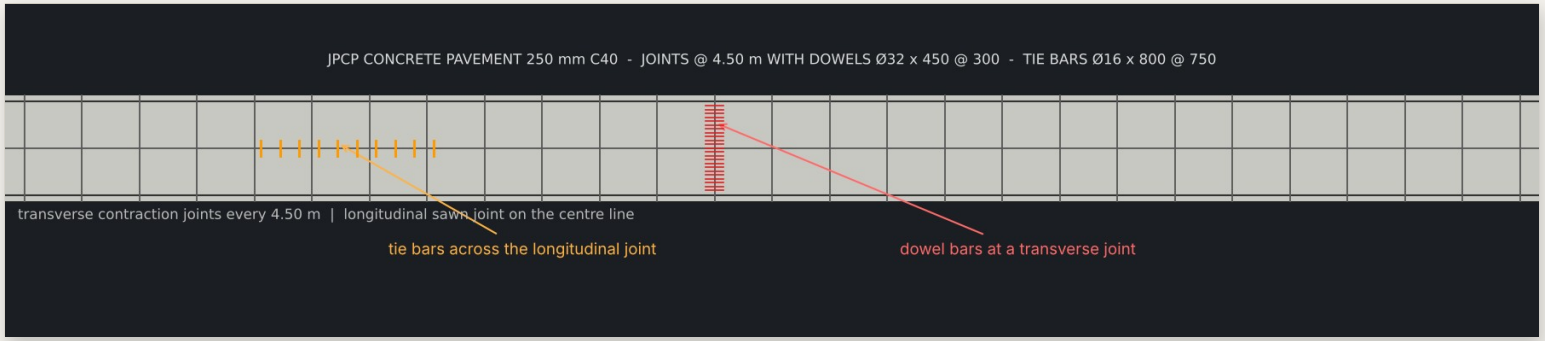


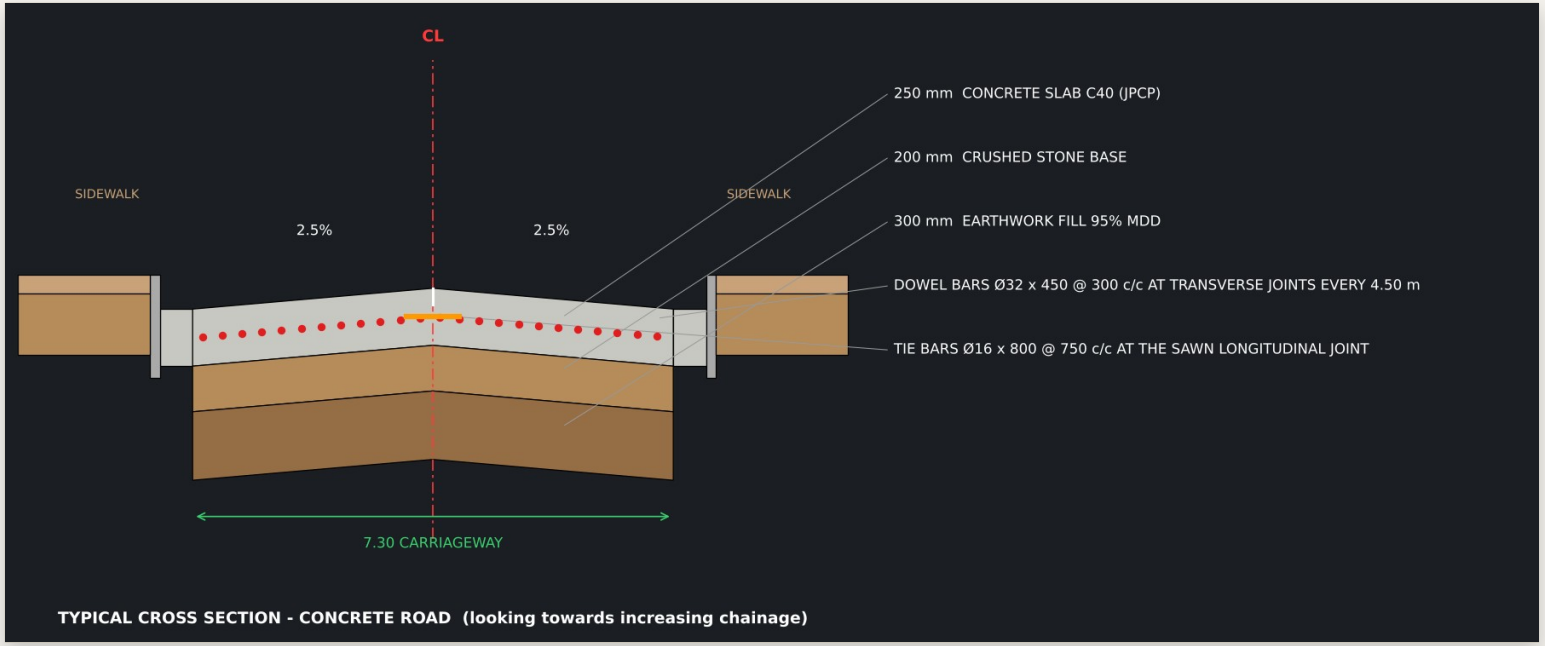
3 Concrete on plan and in section

Joint plan — transverse joints every 4.50 m and the longitudinal joint, on layer ROAD-CONC-JOINT, with a label along the road.

Cross section — slab, stone base and earthwork layers, dowels, tie bars and mesh (ROAD-XSEC-PCC / EARTH / STEEL).

Change it any time with RDPV — the plan, section and report follow.





TIP Concrete can also run over the paved shoulders — tick the option in RDPV.

4 Draw cross sections — RDX

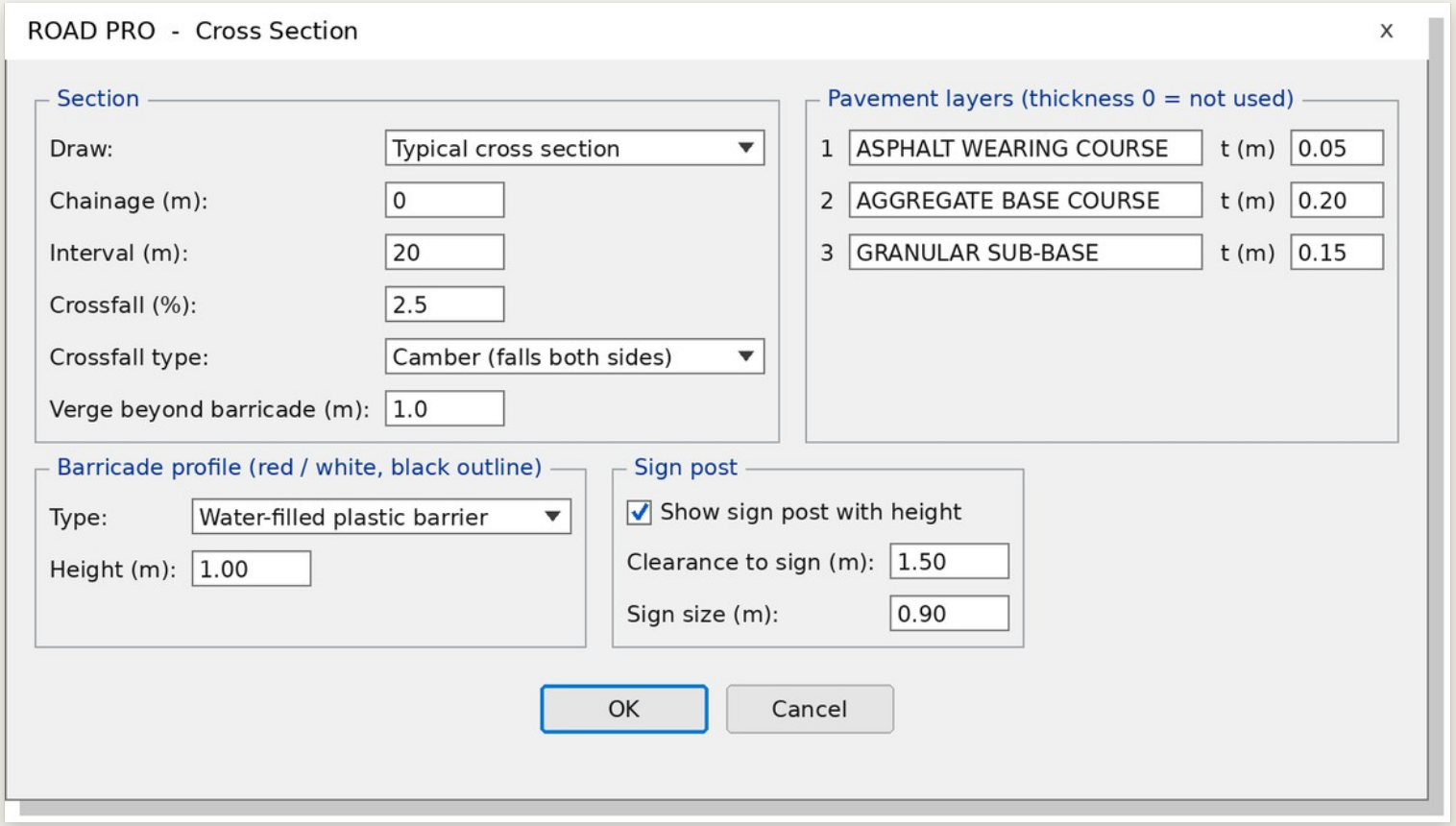
Type RDX and select the road centre line.

Draw: a typical section, a section at every chainage interval, or at one chainage.

Crossfall: percentage and type (camber, one way).

Pavement layers for asphalt roads; concrete roads draw the slab, base, earthwork and steel automatically.

Barricade profile and sign post with height and clearance.



5 Get the quantities — RDR

CONCRETE PAVEMENT (NEAT QUANTITIES - ADD WASTAGE)	
Pavement type	JPCP - jointed plain (dowels + tie bars)
Concrete slab	250 mm C40, area 1660.00 m2, volume 415.00 m3
Stone base	200 mm Crushed stone base, volume 332.00 m3
Earthwork	300 mm Fill 95% MDD, formation 1660.00 m2, volume 498.00 m3
Transverse joints (dowelled)	44 joints @ 4.50 m, total 365.2 m
Longitudinal joints (tied)	1 line(s), total 600.0 m
Dowel bars	dia 32 x 450 mm @ 300 mm : 1188 nos, 3379.2 kg
Tie bars	dia 16 x 800 mm @ 750 mm : 800 nos, 1011.4 kg
TOTAL steel	4.391 t
Joint sealant	965.2 m
Pavement note	Neat quantities from the plan - confirm with the pavement design

1 Type RDR

Select the road centre line and pick the top-left corner for the tables.

2 Read the tables

Road data, curves, corridor, junctions and the pavement table — example: 200 m concrete road, 7.30 m wide.

3 Open in Excel

Answer Yes to open the CSV file with every table for your bill of quantities.

! Neat quantities

No wastage is added — check them against the pavement design.

04

Widening & tapers

Add an extra lane between two chainages, or make the road wider or narrower from a chainage onward.

RDW · RDT

1 Widen a lane — RDW

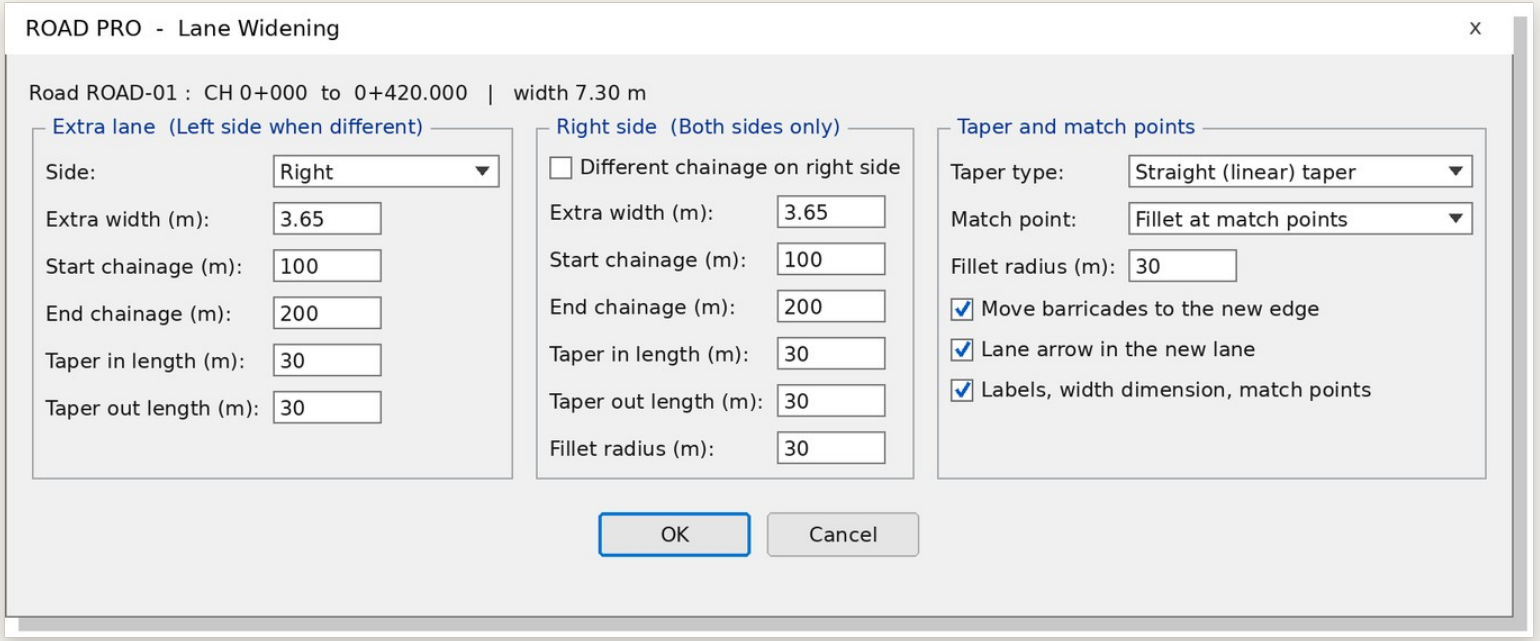
Type RDW and select the road centre line.

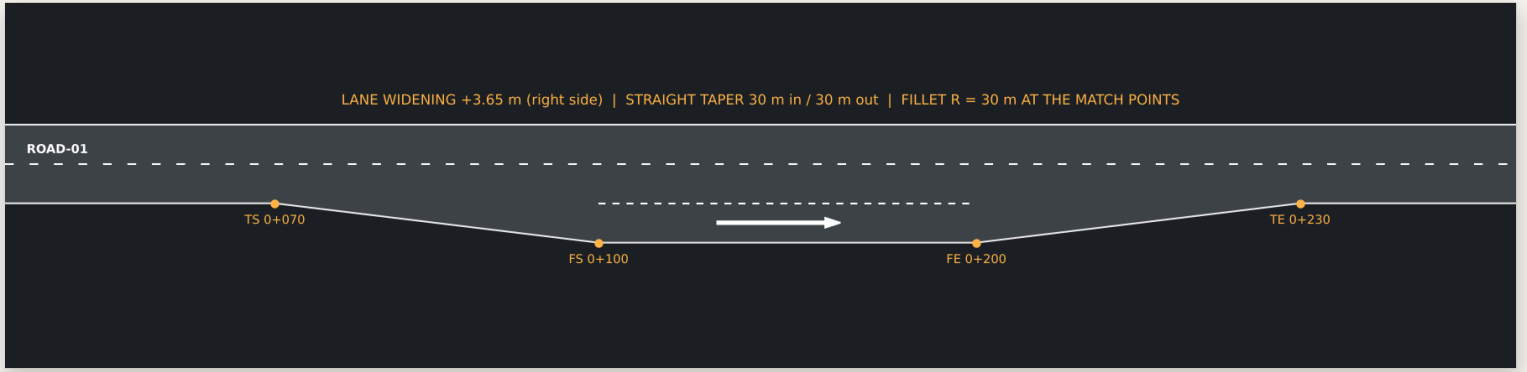
Side and extra width — left, right or both, e.g. +3.65 m.

Start and end chainage of the full width, plus taper in and taper out lengths.

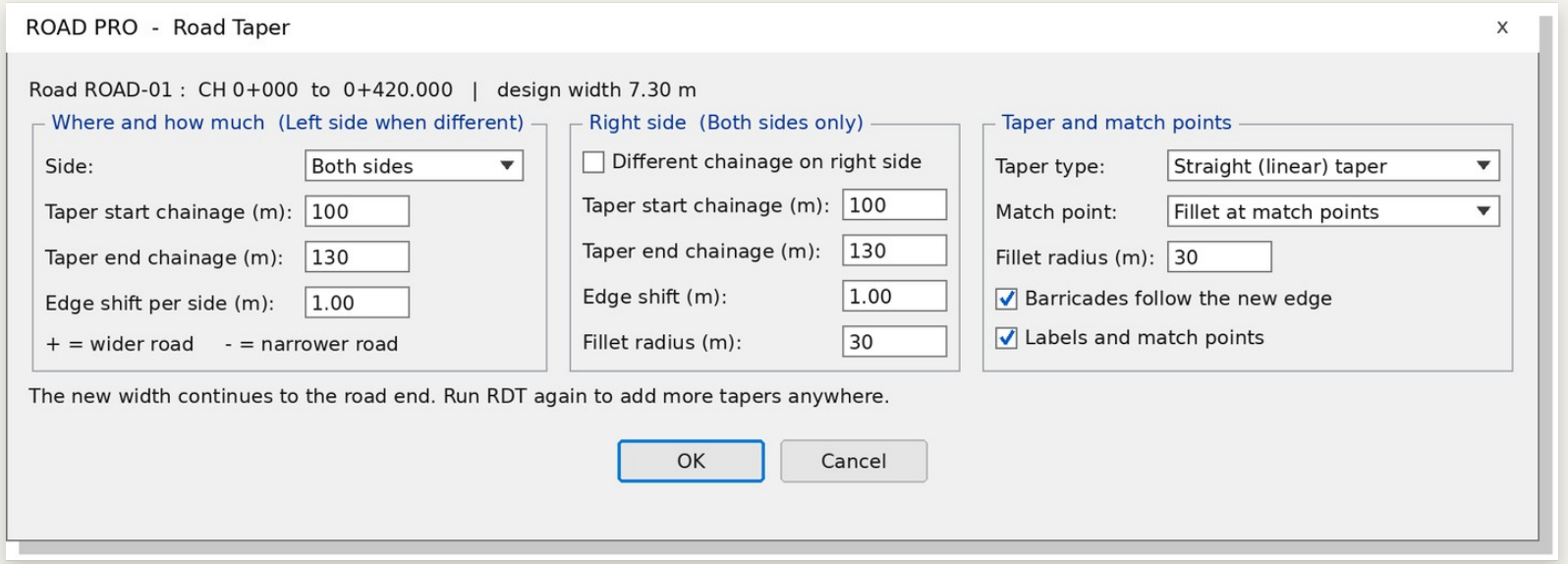
Taper type and match point — straight or curved taper, fillet radius at the match points.

Barricades and arrows move to the new edge; labels show TS, FS, FE and TE.





2 Taper the road wider or narrower — RDT



1 Type RDT

Select the road centre line and choose the side: left, right or both.

2 Taper start and end

Chainages where the edge starts and finishes moving.

3 Edge shift

+ makes the road wider, – makes it narrower. The new width continues to the road end.

! More tapers

Run RDT again to add tapers anywhere. RDB and live update keep them.

05

Junctions & Saudi standard

Kerb returns with the corridor wrapped round the corners, values from the Saudi standard, and sight triangles.

RDJ = ROADJUNCTION

1 Start a junction — RDJ

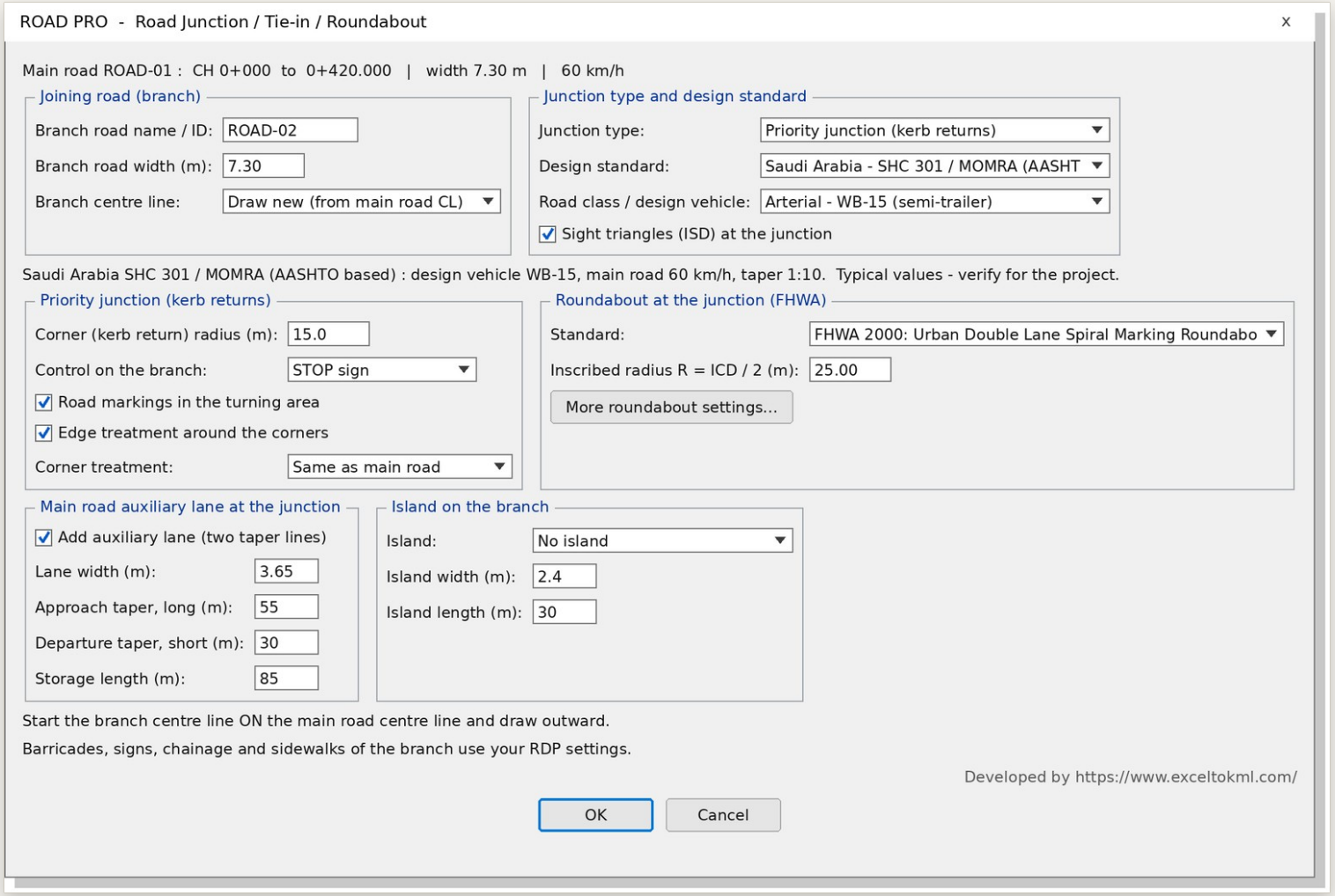
Type RDJ and select the MAIN road centre line.

Joining road: name, width, and draw the branch centre line new — or select one.

Junction type: Priority junction (kerb returns) or Roundabout.

Design standard: Manual, or Saudi Arabia — SHC 301 / MOMRA, then pick the road class / design vehicle.

Control and island: STOP or give way, auxiliary lane, island on the branch.



TIP Start the branch centre line ON the main road centre line and draw outward.

2 Saudi Arabia standard — what RoadPro fills in

Road class	Design vehicle	Corner radius	Aux. lane	Island W	Time gap tg	ISD at 50 / 60 / 80 km/h
Local street	P (car)	9.0 m	3.00 m	1.8 m	7.5 s	104 / 125 / 167 m
Collector	SU (single-unit truck)	12.0 m	3.50 m	1.8 m	9.5 s	132 / 158 / 211 m
Arterial	WB-15 (semi-trailer)	15.0 m	3.65 m	2.4 m	11.5 s	160 / 192 / 256 m
Industrial / heavy	WB-19	18.0 m	3.65 m	2.4 m	11.5 s	160 / 192 / 256 m

Taper ratio 1:8 up to 50 km/h · 1:10 up to 60 · 1:12 up to 70 · 1:15 above

Island length 15 m up to 60 km/h, 30 m above

Storage length deceleration length for the main road speed minus the taper

Sight distance ISD = 0.278 × V × tg, decision point 4.4 m from the edge

NOTE These are typical AASHTO-based values. Always check them against the project standard — every field stays editable.

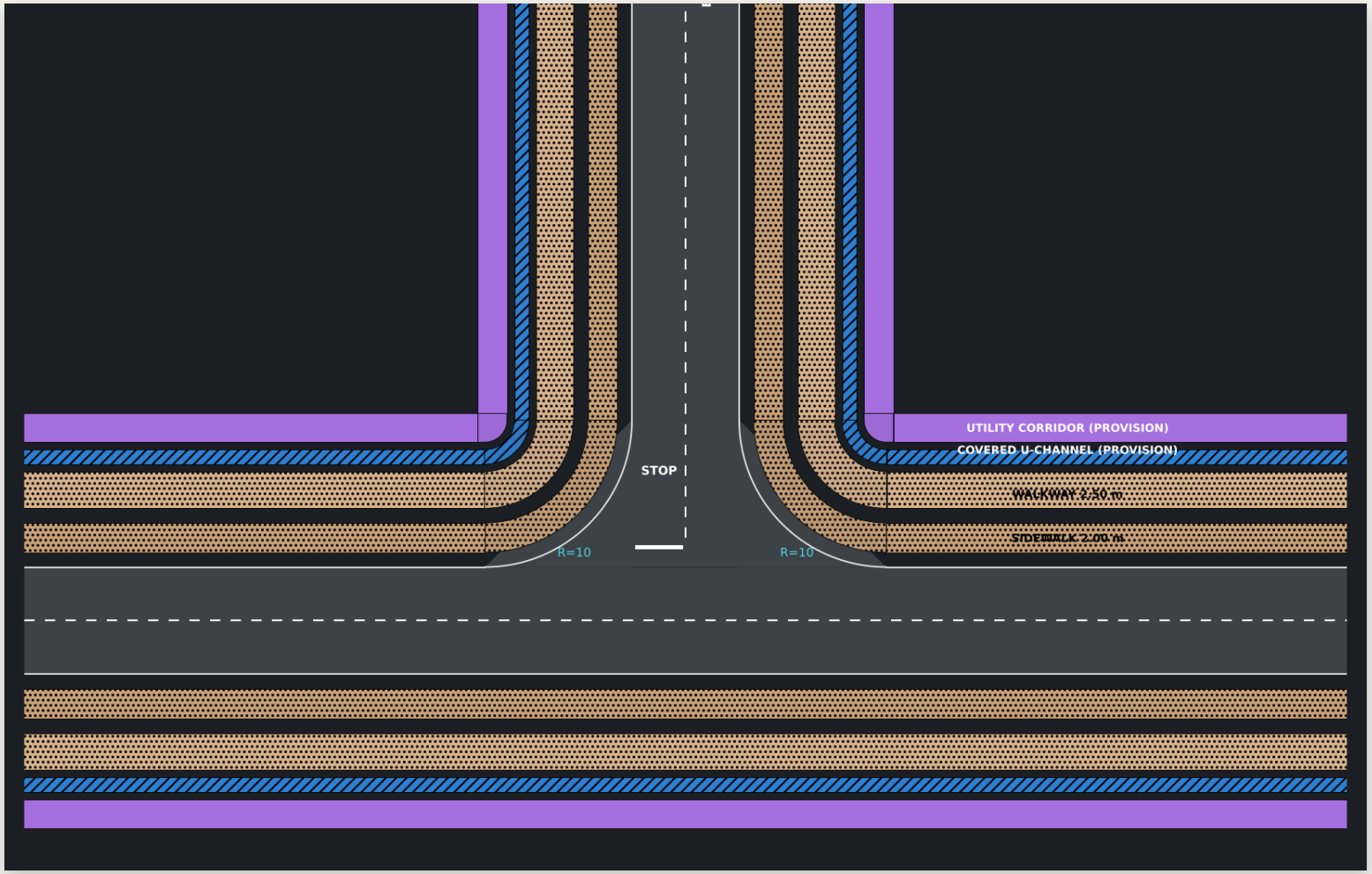
3 Result — the corridor wraps round the corners

Kerb returns are drawn with the corner radius on both sides of the branch.

Sidewalk, walkway, drain and utility strips follow each kerb return with concentric curves and are hatched — no gaps.

The branch utility corridor joins the main road corridor.

STOP line, sign and markings are placed on the branch approach.



TIP Where the corner radius is smaller than the strip offset, the strip turns a sharp corner instead of a curve.

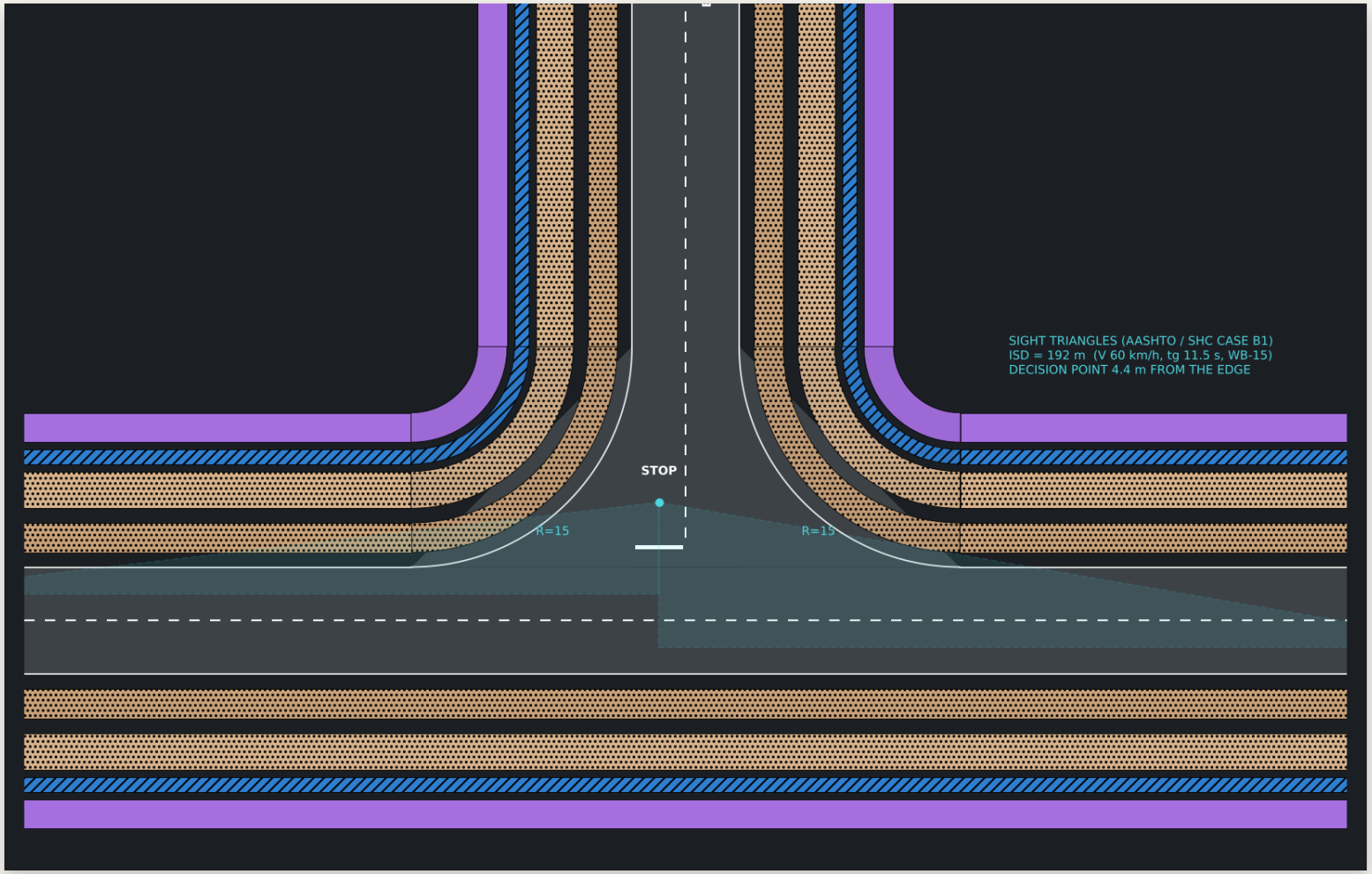
4 Sight triangles at the junction

Tick “Sight triangles (ISD)” in RDJ — it is ticked by the Saudi standard.

Decision point 4.4 m back from the main road edge, in the branch approach lane.

Two triangles: to the near lane on one side and the far lane on the other, at ISD = 0.278 V tg.

Layer ROAD-SIGHT — the ISD is also listed in the report.



TIP Keep the triangles clear of signs, walls and planting higher than about 1 m.

06

Roundabouts

FHWA 2000 and Roundabouts: An Informational Guide 2010 — build one on a road with RDRB, or as the junction type in RDJ.

RDRB = ROADROUNDABOUT

1 Pick a design standard

Standard	Speed	Lanes	ICD	Circ. width	Apron	Entry / exit R
Manual — your values	–	–	–	–	–	–
FHWA 2000: Mini	25	1	13–25 m	5.0	traversable	10 / 15
FHWA 2000: Urban Compact	25	1	25–30 m	5.0	2.0	12 / 15
FHWA 2000: Urban Single Lane	35	1	30–40 m	5.5	2.5	20 / 30
FHWA 2000: Urban Double Lane	40	2	45–55 m	9.5	2.0	30 / 40
FHWA 2000: Urban Double Lane Spiral (default)	**40**	**2**	**45–55 m**	**9.5**	**2.0**	**30 / 40**
FHWA 2000: Rural Single Lane	40	1	35–40 m	5.5	3.0	25 / 40
FHWA 2000: Rural Double Lane	50	2	55–60 m	10.0	3.0	35 / 50
FHWA 2000: Rural Double Lane Spiral	50	2	55–60 m	10.0	3.0	35 / 50
Guide 2010: Mini-Roundabout	30	1	14–27 m	5.0	traversable	10 / 15
Guide 2010: Single-Lane	40	1	27–55 m	5.5	3.0	20 / 30
Guide 2010: Multilane	45	2	46–91 m	9.5	3.0	30 / 45
Guide 2010: Multilane Spiral Marking	45	2	46–91 m	9.5	3.0	30 / 45

Each standard fills
inscribed radius, circulatory width, truck apron, entry and exit radii, splitter islands, lanes and speed — all editable.

Units
speed km/h; widths and radii in metres. ICD = inscribed circle diameter = 2 R.

TIP Traffic keeps right: circulation is anticlockwise.

2 Option A — a roundabout from RDJ

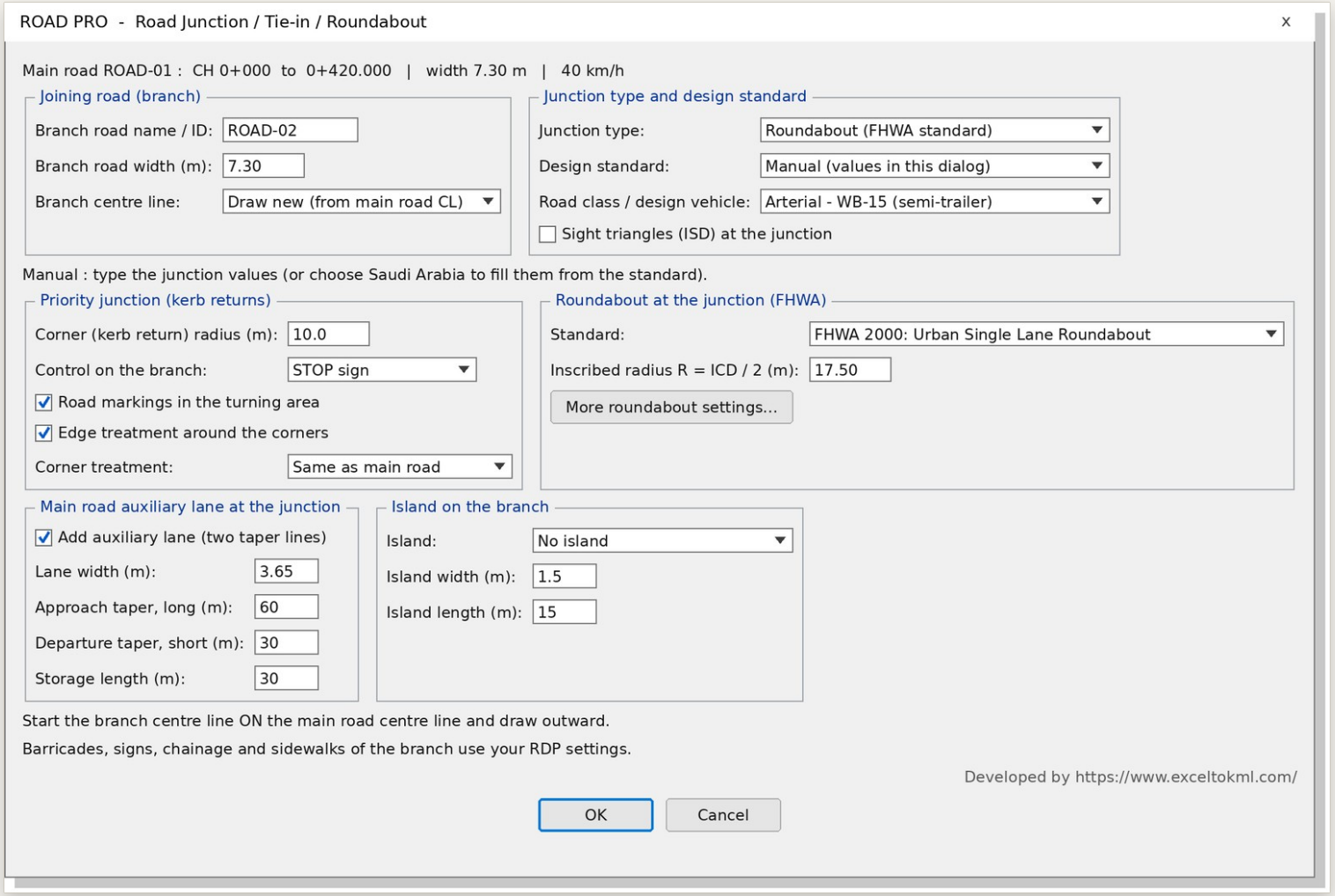
Type RDJ and select the main road.

Junction type = Roundabout. The roundabout panel becomes active.

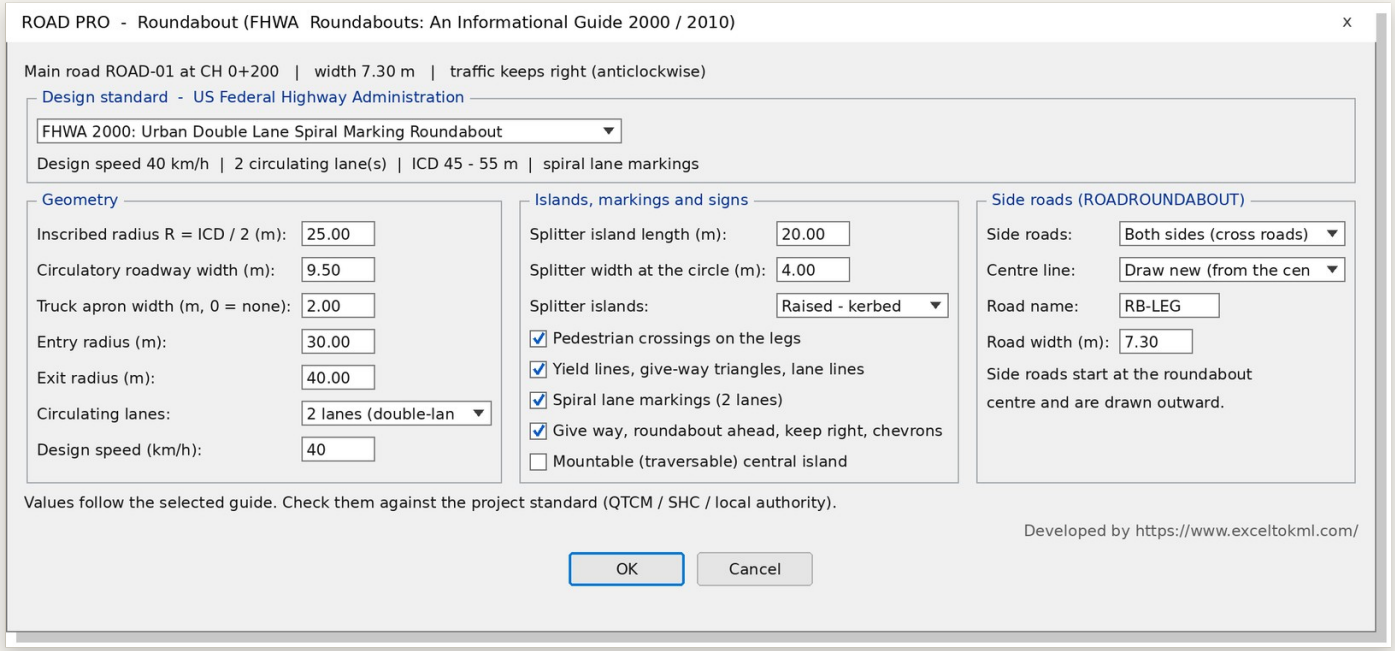
Pick the standard and type the inscribed radius R.

More roundabout settings ... opens the full roundabout dialog (islands, lanes, signs).

Press OK and draw the branch — it becomes a leg of the roundabout.



3 Option B — a roundabout on a road: RDRB



1 Select the main road

Type RDRB and select the MAIN road centre line.

2 Pick the centre

Pick a point on the centre line — or type C and a chainage, e.g. 0+200.

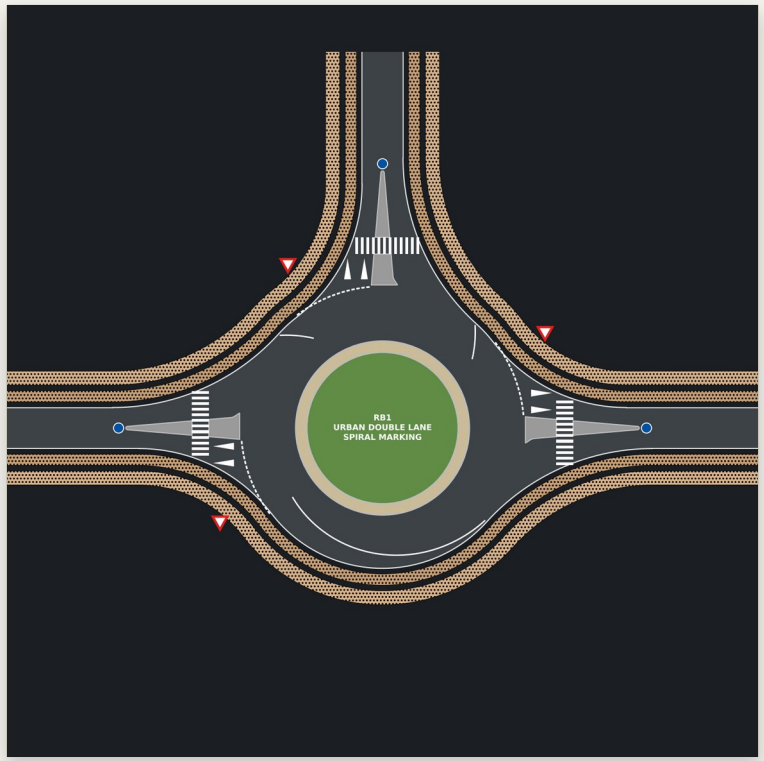
3 Set the dialog

Standard, geometry, islands, markings, signs — and side roads: left, right or both.

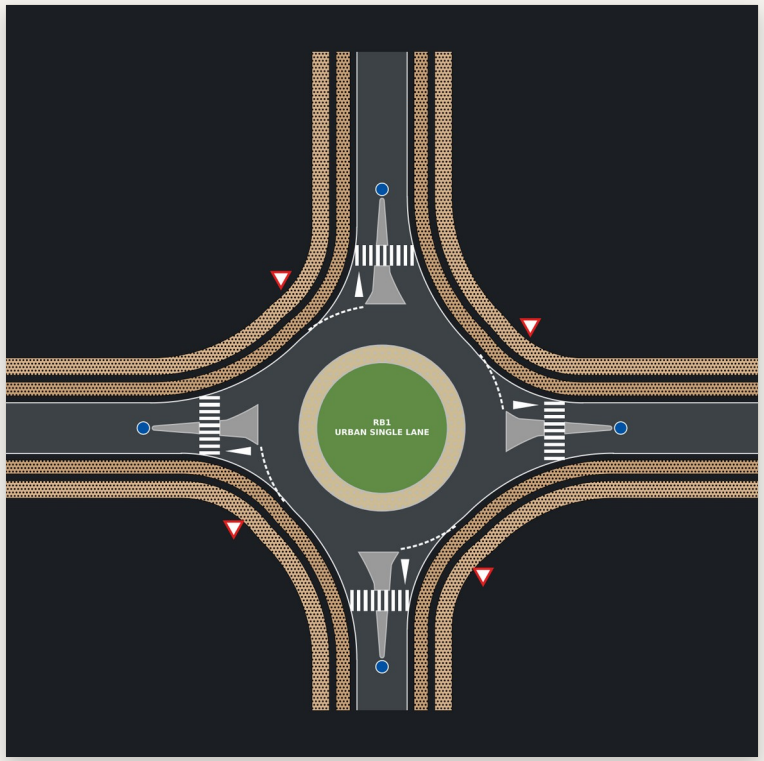
4 Draw the side roads

From the centre outward, or select existing centre lines. They become legs.

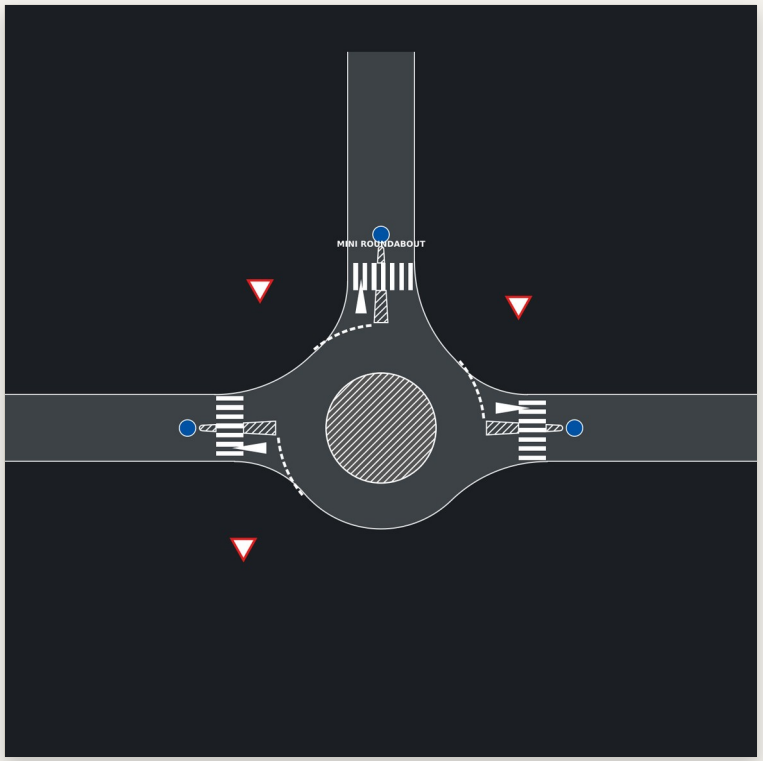
4 What RoadPro draws



Urban double lane, spiral marking — T junction



Urban single lane — cross roads



Mini roundabout — traversable island

Kerb line with entry and exit fillets · circulatory roadway and truck apron · raised or mountable central island · splitter islands that fill the entry flare · pedestrian crossings · yield lines and give-way triangles · concentric or spiral lane line · GIVE WAY, ROUNDABOUT AHEAD, KEEP RIGHT signs and chevrons · ICD dimension, radii, setting-out and report · sidewalks, walkways, drains and utilities wrapped round every corner

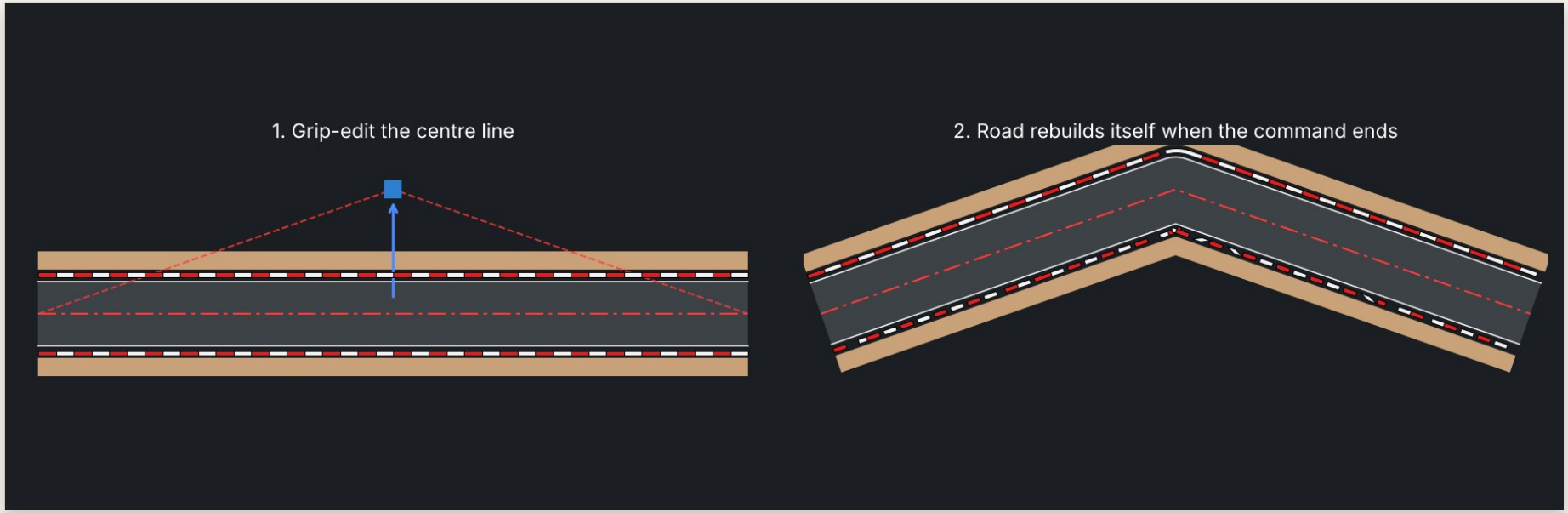
07

Edit & live update

Change a centre line and the whole road follows — widenings, tapers, junctions and roundabouts included.

RDLIVE · RDB · RDER · RDU

1 Live update — edit the centre line, the road follows



1 Live update is ON

Type RDLIVE to switch it On or Off. RDHELP shows the current state.

2 Edit the centre line

Grip-edit, MOVE, ROTATE, STRETCH or PEDIT a road centre line on layer ROAD-CL.

3 Automatic rebuild

When the command ends the road, corridor, markings, widenings, tapers, junctions and roundabouts are rebuilt.

! One U undoes it

Moving a main road also moves the side roads joined to it.

2 Rebuild and edit tools

RDB

Rebuild

Rebuild from the edited centre line. Widenings, tapers, junctions and roundabouts are replayed, not dropped.

RDER

Edit a radius

Select the centre line on a bend and type the new radius — at the centre line or the inner edge — for one bend or all.

RDU

Update

Regenerate the corridor and markings after you change the RDC settings.

ROADPRORESET

Defaults

Restore the default settings (width 7.30 m, radius 10 m) for the next road.

TIP Roads built with v3.4 keep working. Rebuild them once with RDB to get the v3.5 replay and live update.

08

Lines, labels & exchange

Barricade or cone runs on any line, markings, curve labels, straight sections and LandXML for Civil 3D.

RDL · RDM · RDRAD · RDSEC · RDLX · RDLXI

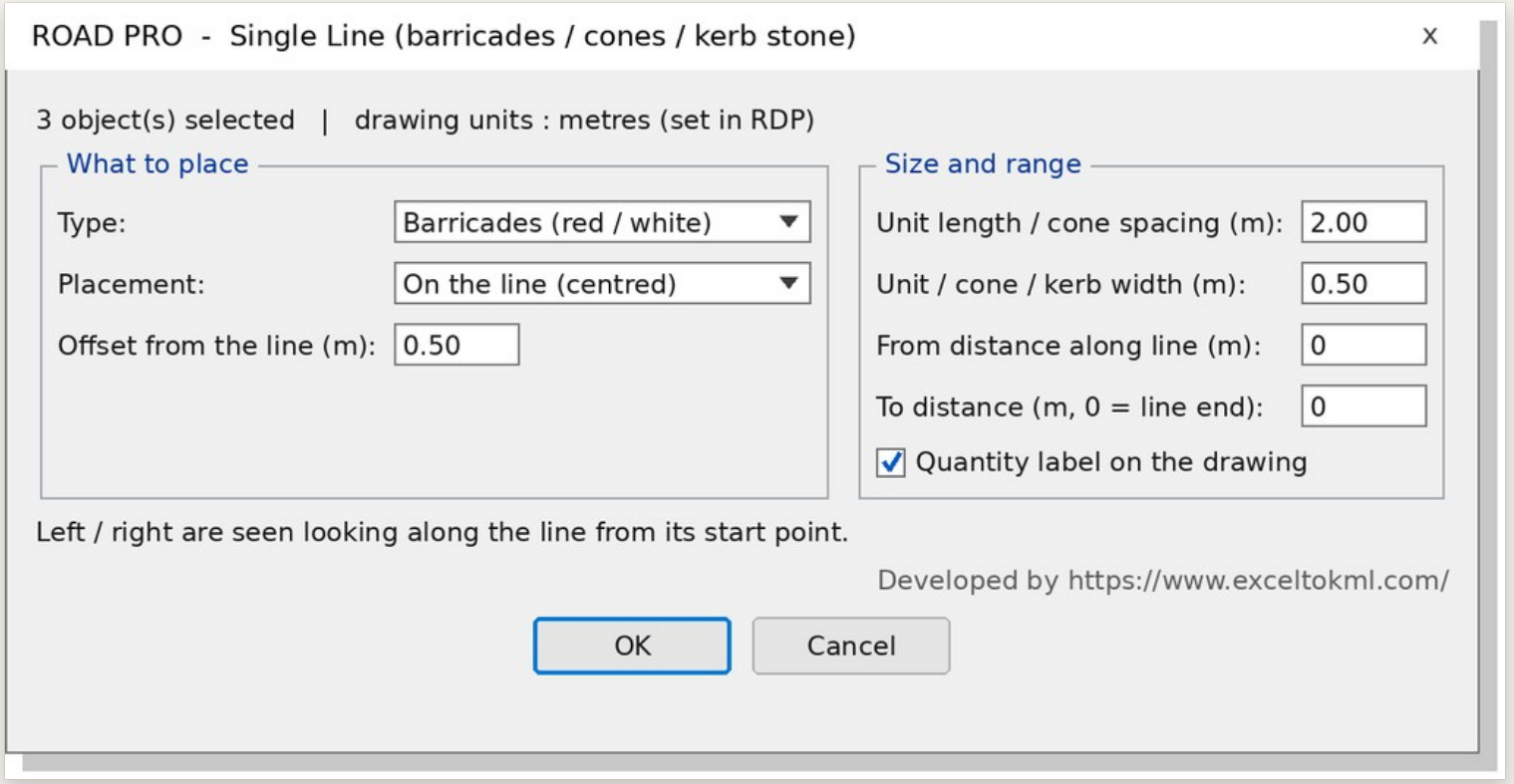
1 Place a single line of barricades, cones or kerb — RDL

Type RDL and select lines, polylines, arcs or splines.

What to place: barricades, cones, kerb stone …, centred on the line or offset left / right.

Size and range: unit length / spacing, width, from and to distance along the line.

Quantity label can be written on the drawing.



TIP Left and right are seen looking along the line from its start point.

2 Markings, labels and LandXML

RDM

Road markings

Draw or remove the road markings of a built road.

RDRAD

Curve data

Radius, TC / CT and curve data labels on plan for every bend.

RDSEC

Straight sections

Label the straights S1, S2 … with their lengths.

RDLX / RDLXI

LandXML

Export a centre line to LandXML for Civil 3D, or import one to build from.

09

Reference

Every command on one page, the new layers and the answers to the most common questions.

RDHELP

1 Command reference

Shortcut	Command	What it does
RDP	ROADPRO	Build a complete road
RDC	ROADCORRIDOR	Lanes, median, sidewalks, drains, utilities
RDPV	ROADPAVEMENT	Asphalt or concrete pavement ★
RDX	ROADXSEC	Cross sections
RDR	ROADREPORT	Report tables + Excel (CSV)
RDW	ROADWIDEN	Lane widening
RDT	ROADTAPER	Road taper
RDJ	ROADJUNCTION	Junction, Saudi standard, roundabout
RDRB	ROADROUNDABOUT	Roundabout on a road ★
RDM	ROADMARK	Road markings
RDL	ROADLINE	Single line of barricades / cones / kerb

Shortcut	Command	What it does
RDRAD	ROADRADIUS	Radius and curve data
RDSEC	ROADSECTIONS	Straight sections S1, S2 …
RDU	ROADUPDATE	Regenerate corridor + markings
RDER	ROADEDITRADIUS	Edit a curve radius
RDB	ROADREBUILD	Rebuild from the edited centre line
RDLIVE	ROADLIVE	Live update on / off ★
RDLX	ROADLANDXML	LandXML export
RDLXI	ROADLANDXMLIN	LandXML import
RDINSTALL	ROADPROINSTALL	Install permanently ★
RDUNINSTALL	ROADPROUNINSTALL	Remove from the Startup Suite ★
RDHELP	ROADHELP	List all commands ★

★ new in v3.5

2 Tips and troubleshooting

RoadPro did not load

Run Install_RoadPro.bat again with AutoCAD closed, or APPLOAD RoadPro.lsp once. On the security prompt choose “Always load”.

“No RoadPro data on this object”

Select the centre line on layer ROAD-CL of a road built by RoadPro — not an edge or a marking.

Edit did not update the road

Check RDLIVE is ON. For a v3.4 road run RDB once. Use U to undo a live update.

Junction cancelled

The branch centre line must start ON the main road centre line and go outward.

Civil 3D alignments

Select an alignment as the centre line in RDP, or exchange centre lines with RDLX / RDLXI.

New layers in v3.5

ROAD-ROUNDABOUT, ROAD-RB-ISLAND, ROAD-RB-APRON, ROAD-SIGHT, ROAD-CONC-JOINT, ROAD-XSEC-PCC, -EARTH, -STEEL.

Build roads faster.

RoadPro v3.5 for AutoCAD and Civil 3D

In the package

RoadPro_v3.5_AddIn.zip — installer and add-in bundle

RoadPro.lsp and RP_PAVER.pat — for APPLOAD

RoadPro_Shortcuts.xlsx — all shortcuts and v3.5 standards

RoadPro_v3.5_WhatsNew.txt — release notes

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